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Python premium

Ds questions

1. Write a python program to copy a data from one variable to other. [ch-3]

```
x = 10
y = x
print(x,y)
```

2. Write a python program to swap two numbers ? [ch-3]

```
x = 50
y = 40
z = x
x = y
y = z
print(x,y)
```

3. Write a python program to format dynamic data and static data ? [ch-3]

```
name = "saurav"
roll = 12
subject = "computer"
print("my name is",name)
print("my roll is",roll)
print("my subject is",subject)
```

4. Write a python program to check the length validity of password ? [ch-6]

```
def checkPassword() :
    password = input("Enter atleast 8 digit password\n")
    length = len(password)
    if length >= 8 :
        print("Login success")
    else :
        print("Login failed !")

checkPassword()
```

5. Write a python program to find the length of a string [ch-7]

```
def findLength() :
    string = "saurav"
    length = len(string)
    print("Length is",length)

findLength()
```

6. Write a python program to convert string in lowercase [ch-7]

```
def convertToLowerCase() :
    string = "SAURAV"
    lowercase = string.lower()
    print(lowercase)

convertToLowerCase()
```

7. Write a python program to convert string in uppercase [ch-7]

```
def convertToUpperCase() :
    string = "saurav"
    uppercase = string.upper()
    print(uppercase)

convertToUpperCase()
```

8. Write a python program to print the first character of a string [ch-7]

```
def printFirstCharacter() :
    string = "SAURAV"
    firstChar = string[0]
    print(firstChar)

printFirstCharacter()
```

9. Write a python program to print the last character of a string [ch-7]

```
def printLastCharacter() :
        string = "SAURAV"
        length = len(string)
        print(string[length-1])

printLastCharacter()
```

10. Write a python program to understand call by value or function call by passing values [ch-7 part-2]

```
def Add(x,y) :
    z = x+y
    return z

def App() :
    x = 10
    y = 12
    result = Add(x,y)
    print("Result is",result)

App()
```

11. Write a python program to understand optional parameters [ch-7 part-2]

```
#Here x=0,y=0 is optional parameters
def Add(x=0,y=0) :
    z = x+y
    return z

def App() :
    x = 10
    y = 12
    result = Add(x,y)
    print("Result is",result)

App()
```

12. Write a python program to call a function multiple times and produce different types of result [ch-7 part-2]

```
def Add(x,y) :
    z = x+y
    return z

def App() :
    result1 = Add(10,12)
    result2 = Add(56,1542)
    result3 = Add(110,121)
    result4 = Add(70,102)
    print("Result no 1",result1)
    print("Result no 2",result2)
    print("Result no 3",result3)
    print("Result no 4",result4)

App()
```

13. Write a python program to understand recursion [ch-7 part-2]

Recursion means : a function that calls itself

```
index = 0
def App() :
    global index
    index = index+1
    print(index,"Executed !")
    if index < 5 : App() #<--Here is recursion logic
    else : print("Recursion end !")

App()
```

14. Write a python program to print the table of user input number using recursion [ch-7 part-2]

```
index = 0

def Table(number) :
    global index
    index = index+1
    print(number,"*",index,"=",number*index)
    if index < 10 : Table(number)
    else : print("Recursion End Or Say Thank You !")

def App() :
    number = int(input("Enter a number\n"))
    Table(number)

App()
```

15. Write a python program to create a list [ch-8]

```
def myList() :
    data = ["saurav","ravi"]
    print(data)

myList()
```

16. Write a python program to create a list and print with index [ch-8]

```
def myList() :
    data = ["saurav","ravi"]
    print(data[0])
    print(data[1])

myList()
```

17. Write a python program to print all the data from list using recursion [ch-8]

```
index = 0
def myList() :
    global index
    data = ["saurav","ravi"]
    print(data[index])
    index = index+1
    length = len(data)
    if index < length : myList()

myList()
```

18. Write a python program to find the index of an element or item using the list [ch-8]

```
index = 0
def findIndex(name) :
    global index
    data = ["saurav","ravi"]
    username = data[index]
    if username.lower() == name.lower()  :
```

```
        print("Matched",name,"at",index,"place")
    index = index+1
    length = len(data)
    if index < length : findIndex(name)

def App() :
    userInput = input("Enter name\n")
    findIndex(userInput)

App()
```

19. Write a python program to remove white space from user input [ch-8]

```
def App() :
    userInput = input("Enter something\n")
    removedWhiteSpace = userInput.strip()
    print(removedWhiteSpace)

App()
```

20. Write a python program to find the largest number from an array or list [ch-10]

```
def max(arr) :
    large = 0
    for num in arr :
        if num > large :
            large = num

    return large

max([1,5,9,8])
```

21. Write a python program to find the smallest number from an array or list [ch-10]

```
def min(arr) :
    small = arr[0]
    for num in arr :
        if num < small :
            small = num

    return small

min([8,9,1,2])
```

22. Write a python program to understand oops concept [ch-15]

```
class Main :
    def __init__(self) :
        print("My first oops program")

Main()
```

23. Write a python program to understand static or class variable in oops [ch-15]

```
class Main :
    x = "i am static or class based variable"
    def __init__(self) :
        print(self.x)

Main()
```

24. Write a python program to understand instance variable in oops [ch-15]

```
class Main :
    def __init__(self) :
        self.x = "Hi, i am creating with the help of instance"
        print(self.x)

Main()
```

25. Write a python program to understand local variable in oops [ch-15]

```
class Main :
    def __init__(self) :
        x = "I am local now"
        print(x)

Main()
```

26. Write a python program to understand the inheritance in oops [ch-15]

```
class Parent :
    def data(self) : print("I am parent class")

class Child(Parent) :
    def __init__(self) :
        super().__init__()
        self.data()

Child()
```

27. Write a python program to understand the encapsulation in oops [ch-15]

```
class Main :
    def __init__(self) :
        self.__secured = "i am private now"

Main()
```

28. Write a python program to understand the polymorphism in oops [ch-16]

```
class Main :
    def __init__(self) :
        self.__secured = "i am private now"

    # setter or mutator
    def setSecured(self,secret) :
        self.__secured = secret

    # getter or accessor
    def getSecured(self) :
```

```
        return self.__secured

main = Main()
main.setSecured("new secret")
print(main.getSecured())
```

29. Write a python program to understand the data abstraction in oops [ch-16]

convert.py

```
class Convert :
    def __init__(self) :
       self.upperCase = lambda string : string.upper()
       self.lowerCase = lambda string : string.lower()
       self.sentenceCase = lambda string : string.title()
```

main.py

```
from lib.convert import Convert
```

```
class Main(Convert) :
    def __init__(self) :
        super().__init__()
        print(self.upperCase("saurav"))
        print(self.lowerCase("SAURAV"))
        print(self.sentenceCase("i am a boy"))

Main()
```

30. Write a python program that checks if a given number is positive, negative, or zero. If the number is positive, Check the number is even or odd. If the number is negative, print "Negative number is not allowed" If the number is zero, print "0 is not acceptable". [ch-19 Part-1]

```
class EvenOddWithException :
    def __init__(self) :
        try : num = int(input("Enter a number\n"))
        except : print("Invalid number")
        else :
            if num > 0 :
                if num % 2 == 0 : print(num,"is Even")
                else : print(num,"is Odd")

            elif num == 0 : print("0 is not acceptable")
            else : print("Negative number is not allowed")

EvenOddWithException()
```

31. Write a python program that takes in a person's age and checks if they are a child (0-12), teenager (13-19), young adult (20-39), middle-aged adult (40-59), or senior (60+). Print an appropriate message for each age group. [Ch-19 Part -1]

```
def personAge() :
    try : age = int(input("Enter your age\n"))
    except : print("Invalid age")
    else :
        if age < 0 : print("Invalid age")
        elif age <= 12 : print("You are in child age")
        elif age <= 19 : print("You are a teenager")
        elif age <= 39 : print("You are adult")
        elif age <= 59 : print("You are in middle age")
        else : print("You are in senior age")

personAge()
```

32. Write a python program to calculate BMI (Body Mass Index)

```
def BMI() :
    try :
        height = float(input("Enter your height in meter\n"))
        weight = float(input("Enter your wight in kilograms\n"))
```

```
    except ValueError as msg : print(msg)
    else :
        bmi = weight / (height*height)
        if bmi < 18.5 : print("Underweight")
        elif bmi < 24.9 : print("Normal weight")
        elif bmi < 29.9 : print("Overweight")
        else : print("Obese")
BMI()
```

33. Write a program that takes in three numbers and checks if they form a Pythagorean triples. If they do, print "These numbers form a Pythagorean triple." If they don't, print "These numbers do not form a Pythagorean triple." [Ch-19 Part -1]

```
def triples() :
    try :
        a = int(input("Enter 1st number\n"))
        b = int(input("Enter 2nd number\n"))
        c = int(input("Enter 3rd number\n"))
    except ValueError as msg : print(msg)
    else :
            if (a*a) + (b*b) == (c*c) :
                 print("These numbers form a Pythagorean triples")
            else :
                 print("These numbers do not form a Pythagorean
triples")

triples()
```

34. Write a Python program that takes in a string and prints whether it contains a lowercase letter, uppercase letter or both. [Ch-19 Part -1]

```
def case() :
    string = input("Enter your string\n")
    capital = any(char.isupper() for char in string)
    small = any(char.islower() for char in string)
    if capital == True and small == True :
        print("String contains both lowercase and uppercase")
    else :
```

```
        if capital == True : print("String contains only
uppercase")
        else : print("String contains only lowercase")

case()
```

35. Write a Python program that asks the user for their age and gender, and prints a message indicating whether they are eligible for a discount. Females under the age of 25 and males under the age of 22 are eligible for a discount. [Ch-19 Part -1]

```
def discount() :
    try :
        age = int(input("Enter you age\n"))
        gender = input("Enter your gender male or female\n")
        if gender.lower() != "male" and gender.lower() != "female"
:
            print("Invalid gender")

    except: print("Invalid age")
    else :
        if gender.lower() == "female" and age < 25 :
            print("Eligible for discount !")

        elif gender.lower() == "male" and age < 22 :
            print("Eligible for discount !")

        else : print("Not eligible for discount")

discount()
```

36. Write a program that takes a number from the user and checks if it's a prime number [Ch-19 Part -2]

```
def primeNumber() :
    try : num = int(input("Enter a number\n"))
    except : print("Invalid number")
    else :
        if num > 1 :
```

```
            for index in range(2,num) :
                if num % index == 0 :
                    print(num,"is not a prime number")
                    break
            else: print(num,"is prime number")
        else : print(num,"is not a prime number")

primeNumber()
```

37. Write a program that takes a list of numbers and returns the sum of all the even numbers in the list. [Ch-19 Part -2]

```
def addEven() :
    sum = 0
    nums = [1,2,3,4,5,6,7,8,9]
    for num in nums :
        if num % 2 == 0 : sum = sum+num
    print("Total is",sum)

addEven()
```

38. Write a program that takes a list of strings and returns the longest string in the list. [Ch-19 Part -2]

```
def longString() :
    long = ''
    strings = ["saurav","ravi","ram","wap institute"]
    for string in strings :
        length = len(string)
        longLength = len(long)
        if length > longLength :
            long = string
    print("Longest string is",long)

longString()
```

39. Write a program that takes a string and returns the reverse of the string. [Ch-19 Part -2]

```python
def reverseString() :
    result = ''
    string = input("Enter your string\n")
    length = len(string)-1
    for index in range(length,-1,-1) :
        result = result + string[index]
    print("Reverse of",string,"is",result)


reverseString()
```

40. Write a program that takes a list of strings and returns a new list with only the strings that contain the letter input by user. [Ch-19 Part -2]

```python
def charSearch() :
    result = []
    strings = ["saurav","sikha","rahul"]
    char = input("Enter a letter\n")
    for string in strings :
        if char in string :
            result.append(string)
    print(result)

charSearch()
```

41. Write a program that takes a string and returns the number of vowels in the string. [Ch-20 Part-1]

```python
def vowelCount() :
    count = 0
    vowels = "aeiou"
    string = input("Enter your string\n")
    for char in string :
        if char.lower() in vowels :
            count = count+1
    print(count)
```

42. Write a program that takes a list of numbers and returns the largest number in the list. [Ch-20 Part 1]

```
def largestNumber() :
    max = 0
    nums = [1,5,9,89,2,63,9]
    for num in nums :
        if num > max :
            max = num
    print("Largest number is",max)

largestNumber()
```

43. Write a program that takes a list of numbers and returns the smallest number in the list. [Ch-20 Part 1]

```
def smallestNumber() :
    nums = [118551,-1,-2,1,8,6,9,3]
    min = nums[0]
    for num in nums :
        if num < min :
            min = num
    print("Smallest number is",min)
```

44. Write a program that takes a list of numbers and returns the second largest number in the list. [Ch-20 Part 1]

```
def secondLargestNumber() :
    nums = [1,9,6,3,8,5]
    large = max(nums)
    sl = nums[0]
    for num in nums :
        if num > sl and num < large :
            sl = num
    print("Second largest number is",sl)

secondLargestNumber()
```

45. Write a program that takes a string and returns True if the string is a palindrome and False otherwise. [Ch-20 Part 1]

```
def palindrome() :
    string = input("Enter your string\n")
    reverse = string[::-1]
    if string == reverse :
        print("String is palindrome")
    else : print("String is not palindrome")

palindrome()
```

46. Write a program that takes a list of numbers and returns a new list with only the prime numbers in the list. [Ch-20 Part 1]

```
def primeList() :
    primeList = []
    nums = [1,2,34,5,6,856,9,6,3,23,58,69,36,30,83]
    for num in nums :
        if num > 1 :
            for index in range(2,num) :
                if num % index == 0 : break
            else : primeList.append(num)
    print("Prime list",primeList)

primeList()
```

47. Write a program that takes a list of numbers and returns a new list with only the numbers that are divisible by 3. [Ch-20 Part 1]

```
def divisible() :
    divisibleByThree = []
    nums = []
    for index in range(10) :
        try : num = int(input("Enter number\n"))
        except: raise Exception("Invalid number")
        else :
            nums.append(num)
    for n in nums :
```

```
        if n % 3 == 0 :
            divisibleByThree.append(n)
    print("These numbers are divisible by 3",divisibleByThree)

divisible()
```

48. Write a program that takes a list of strings and returns a new list with all the strings in uppercase. [Ch-20 Part 1]

```
def upperList() :
    uppers = []
    strings = ["saurav","ravi","ranjan","sikha"]
    for string in strings :
        uppers.append(string.upper())
    print("Lowecase list",strings)
    print("Uppercase list",uppers)

upperList()
```

49. Write a program that takes a list of numbers and returns a new list with only the numbers that are odd. [Ch-20 Part 1]

```
def oddList() :
    odds = []
    nums = [1,5,9,24,2,5,251,65,56,92,3]
    for num in nums :
        if num % 2 != 0 :
            odds.append(num)
    print("Odd list",odds)

oddList()
```

50. Write a program that takes a string and returns the number of words in the string. [Ch-20 Part 1]

```
def wordCount() :
    string = input("Enter your sentence\n")
    stringList = string.split()
    count = len(stringList)
```

```
    print(count)

wordCount()
```

51. Write a program that takes a list of strings and returns a new list with all the strings that start with the letter 'a'. [Ch-20 Part 1]

```
def findLetter() :
    result = []
    strings = ["wap","good","apple","any","hello"]
    for string in strings :
        if string.lower().startswith('a') :
            result.append(string)

    print("Start with a",result)

findLetter()
```

52. Write a program that takes a list of numbers and returns a new list with only the numbers that are perfect squares. [Ch-20 Part 2]

```
def perfectSquare() :
    nums = [2,5,16,9,12,36,44,4]
    result = []
    for num in nums :
        if int(num**0.5)**2 == num :
            result.append(num)
    print("These are perfect squares",result)

perfectSquare()
```

53. Write a program that takes a list of strings and returns a new list with all the strings that have more than one word. [Ch-20 Part 2]

```
def moreWord() :
    strings = []
    result = []
    for index in range(5) :
```

```
        user = input("Enter your string\n")
        strings.append(user)

    for string in strings :
        length = len(string.split())
        if length > 1 : result.append(string)

    print("More than one word",result)

moreWord()
```

54. Write a program that takes a list of numbers and returns a new list with only the numbers that are even & greater than 10. [Ch-20 Part 2]

```
def greaterList() :
    nums = []
    result = []
    for index in range(5) :
        try : num = int(input("Enter your number\n"))
        except: raise Exception("Invalid number")
        else: nums.append(num)

    for num in nums :
        if num % 2 == 0 and num >= 10 : result.append(num)

    print(result)

greaterList()
```

55. Write a program that takes a list of numbers and returns a new list with only the numbers that are less than 5. [Ch-20 Part 2]

```
def smallerList() :
    nums = []
    result = []
    for index in range(5) :
        try : num = int(input("Enter your number\n"))
        except: raise Exception("Invalid number")
```

```
        else: nums.append(num)

    for num in nums :
        if num <= 5 : result.append(num)

    print(result)

smallerList()
```

56. Write a program that takes a list of strings and returns a new list with all the strings that end with the letter 's'. [Ch-20 Part 2]

```
def endWith() :
    strings = ["try","wap","labs","waves","man"]
    result = []
    for string in strings :
        length = len(string)
        if string[length-1].lower() == 's' :
            result.append(string)

    print(result)

endWith()
```

57. Write a program that takes a list of numbers and returns the sum of all the numbers in the list. [Ch-20 Part 2]

```
def listSum() :
    nums = []
    sum = 0
    for index in range(5) :
        try : num = int(input("Enter a number\n"))
        except : raise Exception("Invalid number")
        else :
            nums.append(num)

    for num in nums :
        sum = sum+num
```

```
    print("Sum of",nums,"is",sum)

listSum()
```

58. Write a program that takes a list of strings and returns a new list with all the strings that are at least 5 characters long. [Ch-20 Part 2]

```
def filterString() :
    strings = []
    for index in range(5) :
        user = input("Enter your string\n")
        strings.append(user)

    result = []

    for string in strings :
        length = len(string)
        if length >= 5 :
            result.append(string)
    print(result)

filterString()
```

59. Write a program that takes a string and returns True if the string contains only letters and False otherwise. [Ch-20 Part 2]

```
def checkAlpha() :
    user = input("Enter string\n")
    return user.isalpha()

checkAlpha()
```

60. Write a python program to count the number of digits in a given integers. [Ch-21 Part 1]

```
def count() :
    length = 0
    try : num = int(input("Enter a number\n"))
```

```
    except: print("Invalid number")
    else :
        while num > 0 :
            length = length+1
            num = num//10
        print(length,"Digits")

count()
```

61. Write a python program to find the factorial of given or user input. [Ch-21 Part 1]

```
def factorial() :
    result = 1
    try : num = int(input("Enter your number\n"))
    except: print("Invalid number")
    else :
        """
        for index in range(num,0,-1) :
            result = result*index
        print("Factorial of",num,"is",result)
        """

        while num >= 1 :
            result = result*num
            num = num-1
        print("Factorial of",num,"is",result)

factorial()
```

62. Write a Python program to print the Fibonacci sequence up to a given number using a while loop. [Ch-21 Part 1]

```
def sequence() :
    try : num = int(input("Enter your number\n"))
    except: print("Invalid number")
    else :
       start0 = 0
```

```
        start1 = 1
        while start1 < num :
            result = start0+start1
            print(result)
            start0 = start1
            start1 = result

sequence()
```